

Database Management System

CSIT — 4th Semester — 2079 — Answer Sheet
Subject Code: CSC265

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. What is Database Management System? Explain the DBMS architecture with diagram. What is data independence?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. What is entities and its types? How do we reduce different types of entities and relationships of ER diagram into Tables? Explain with assuming different types of entities and relations.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. What is normalization? Why normalization is required? Explain 1NF, 2NF and 3NF with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

4. What do you mean by Schema and instance in DBMS? Explain both with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. What is conflict schedule? Explain with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Retrieve the TName, and No_of_priod of teachers who teach in "ABC" school using Relational Algebra. TEACHER(TID, TName, TAddress, TQualification) SCHOOL (SID, SName, SAddress, SPhone) SCHOOL_TEACHER (SID, TID, No_of_period)

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Explain aggregation with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Define functional dependencies. Explain 2nd normal form with example?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Explain Assertion and Triggers with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. What is concurrency control? What are its advantages in DBMS?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. What is Buffer Management in DBMS? Explain.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. What is transaction? Draw states of transaction and explain.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.